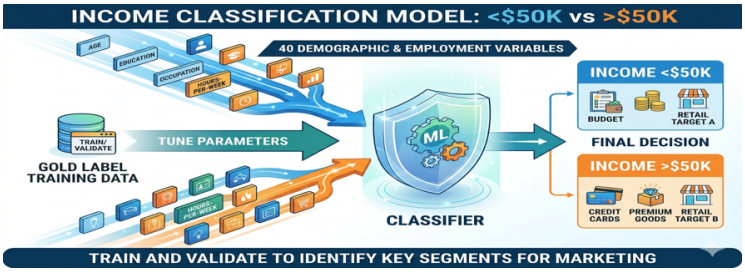
RETAIL RECO: RECOMMENDATION SYSTEM FOR RETAIL MARKETING

How can a retail business identify high-income customers from demographic data, and how can they group their audience into meaningful segments to drive smarter marketing decisions?

1. PROJECT INTRODUCTION

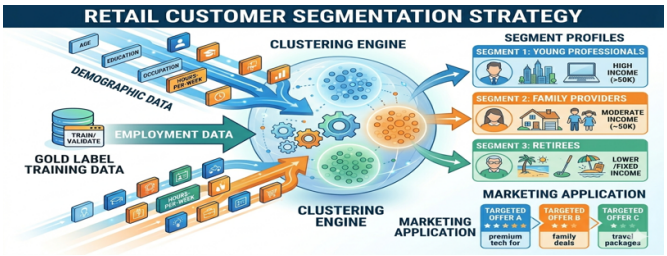
In this project, we aim to build two analytical models for a retail business client to support their marketing efforts. The first model is a classification model that predicts whether a given individual earns more or less than $50,000 per year.

Income Classifier: By identifying high-income individuals from demographic and employment data, the client can target their marketing campaigns more precisely and avoid wasting budget on audiences unlikely to respond.



Spend your marketing budget where it matters most!

Customer Segmenter: The second model is a segmentation model that groups the population into distinct clusters based on shared demographic and employment characteristics. It lets clients avoid generic campaigns and build custom plans for each group - Mainly useful when targeting specific markets (like retirees or tech workers).



Are all your customers same?

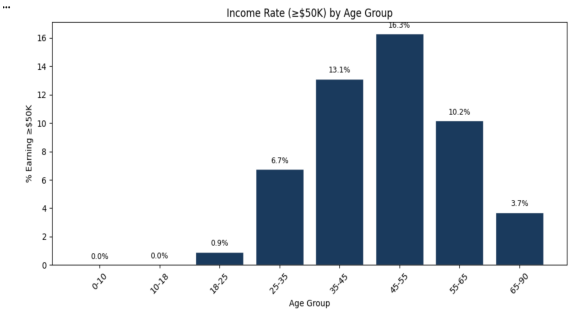
2. DATA EXPLORATION

The data used in this project comes from the 1994 and 1995 U.S. Census Bureau Current Population Surveys, containing 40 demographic and employment variables across 199,523 records. The dataset presented several real-world challenges including severe class imbalance, missing values, duplicate records, and a mix of categorical and numerical features all of which were addressed during preprocessing.

2.1 Getting to Know the Data

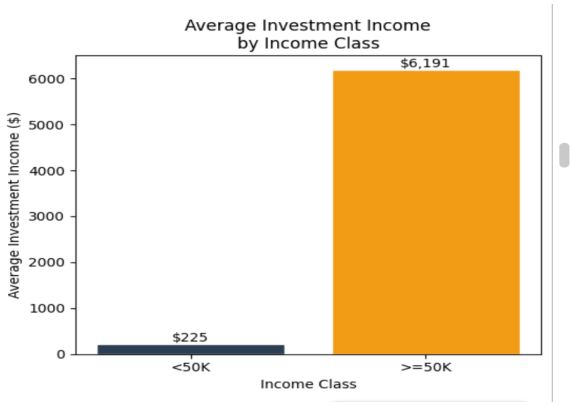
The dataset consists of a wide range of features, among which are age and gender (demographics), capital gains and dividends (income-related), occupation and weeks worked (employment), citizenship of parents, place of birth, education level (background). On the outset, all these variables seem to be good predictors of income. But, including all attributes without a thought is highly unlikely to be a good solution; some features carry real predictive

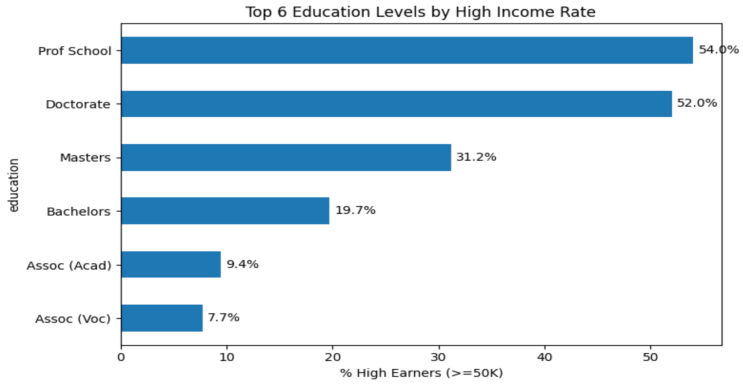
information, whereas others are just distractions for the model that lower its performance. This is the main reason why, along with modeling, excellent feature selection and data handling are just as important.



AGE: The chart shows age groups 35-55 are in the highest bracket of earners (over $50,000) with 16.3% rate of highly paid people. From a retailer's point of view, these mid-career men and women will be the most profitable demographic for income-based marketing campaigns. On the other hand, the youngest and oldest age groups would be less likely to be high earners, and they will probably have to be approached through other marketing methods.

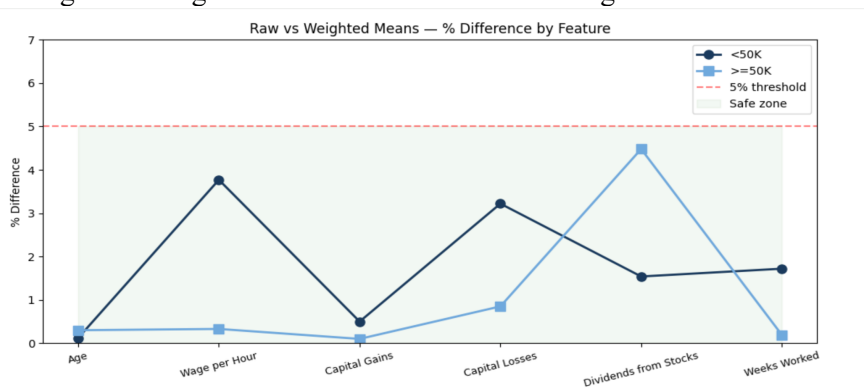
INVESTMENT INCOME: The data shows a significant disparity in investment income between the two groups. People earning more than $50,000 enjoy an average investment income of $6,191, compared to only $225 for those earning less than $50,000, a nearly 28-fold difference. This makes total investment income one of the most powerful indicators for identifying wealthy incomes. Individuals with significant capital gains, dividends, or stock income are more likely to fall into the high-income class and should be emphasized in targeted marketing campaigns.





EDUCATION: Customers with higher education levels, particularly those with professional, doctoral, and master's degrees, are far more likely to make more than $50,000. since a result, the retail industry should prioritize targeting these sectors, since they have the most potential for conversion and revenue production.

WEIGHTS: Each person in the census dataset is associated with a weight assigned to them by the census which indicates how many other people in the population they resemble. We briefly tested whether or not if utilizing or ignoring these weights would affect our understanding of the data.



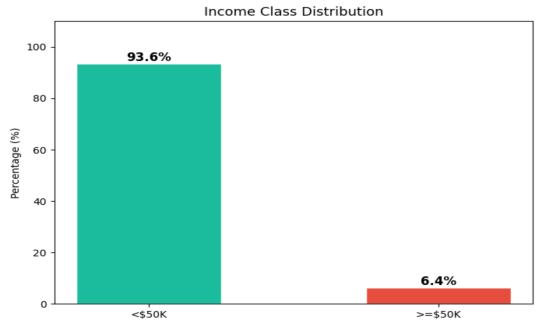
The answer is "no." Every single attribute remains well below the 5% difference criterion for both income groups, implying that the data is largely the same with or without the weights. This provides assurance that deleting the weight column was a sound and justifiable move that did not bring bias into the analysis.

2.2 Dealing with Missing Values: The dataset included missing values in two forms: ordinary nulls and placeholder?, which were standardized into a single format. Four migration-related columns with almost 50% missing values were removed, and the remaining five columns with less than 4% missingness were imputed with the most common value.

2.3 Outlier Detection: Outlier detection indicated no differences in age, weeks worked, or number of people worked for. However, financial columns revealed a tiny fraction of extreme values, most notably stock dividends (10.6%) and hourly wage (5.67%). These were included in the sample because they represent true high earners with large investment and wage income, which is exactly the population the client wants to target.

2.4 Cleaning Up the Labels: The income label column was a bit messy values like '- 50000.' and '50000+.' had extra spaces and dots. Cleaned those up and converted them to simple 0s and 1s. So that its best for the model.

2.5 The Class Imbalance Problem



The dataset is severely skewed, with 94% earning less than $50,000 and only 6% earning more. A model that merely forecasts that everyone has a low income would appear 94% accurate but would be entirely useless for the client's needs. To address this, the model was modified to pay special attention to high earners during training, and evaluation criteria were set to indicate true predictive performance rather than merely total accuracy.

3. PRE-PROCESSING APPROACH

Before creating any model, the data was cleaned and properly formatted. The survey sample weight and year columns were removed because the weight shows population distribution rather than individual attributes, and the year column, which only covers 1994-1995, provides no relevant signal.

3.1 Feature Encoding: Amongst mixed categorical and numerical columns, the dataset had, Categorical columns were converted to LightGBM's native category dtype which made the model understand these features directly without manual encoding. This method doesn't lead to the explosion of feature space due to encoding of high-cardinality columns like major_occupation_code. For the clustering model, categorical columns were encoded numerically since K-Means requires all features to be numerical. The numerical features were normalized by StandardScaler to make sure that a feature's scale alone would not lead it to dominate the distance calculations.

3.2 Train / Validation / Test Split: The samples were split into three groups- training set was 67%, validation was 13% and test was 20%. The splitting was done stratified so that the original 94/6 class proportion gets maintained in the three sets resulting in reliable evaluation.

3.3 Addressing Class Imbalance: The ratio of income classes in the sample is 15:1. Instead of resampling the dataset arbitrarily, LightGBM used scale_pos_weight = 14.84 to give a higher penalty for misclassifying high earners in the training phase. This only would be relevant to the classification model since segmentation model does not have a target label and hence not affected by class imbalance.

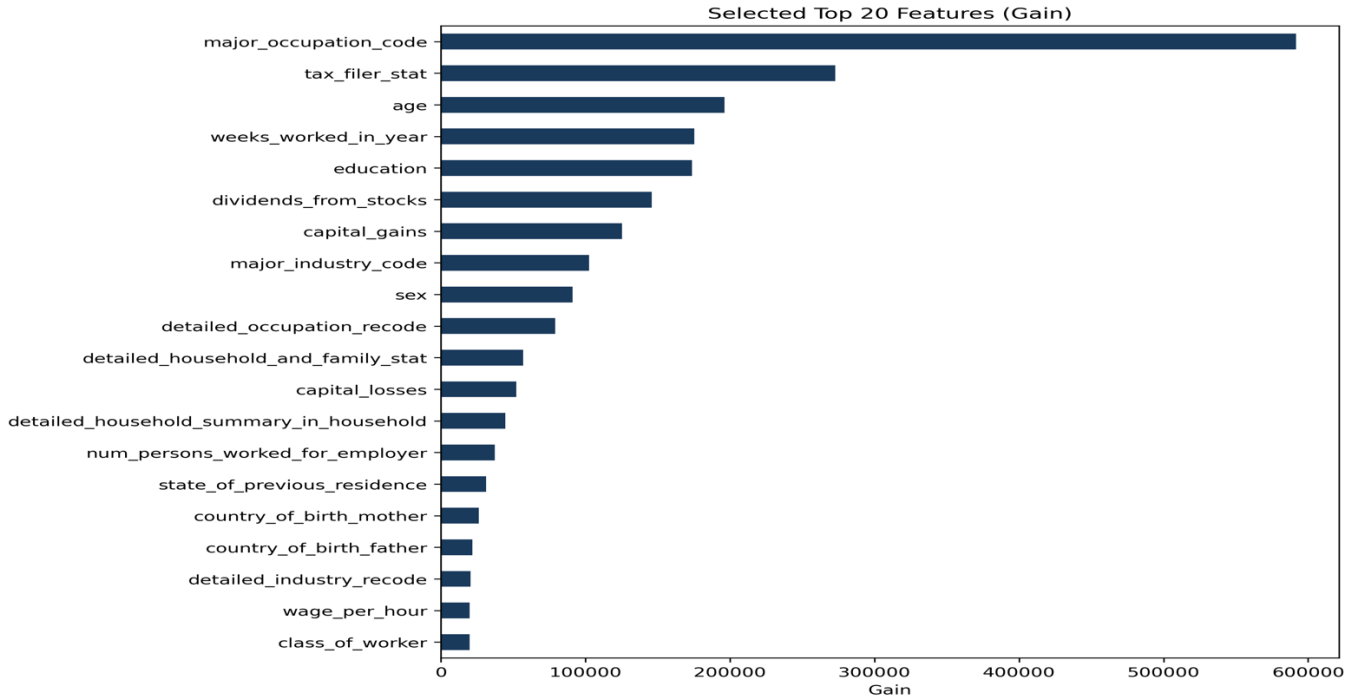
3.4 Threshold Tuning: The precision-recall curve was used to find the best classification threshold, as opposed to the default 0.5 probability threshold. The final threshold of 0.87 was chosen to maximize F1 score, achieving the correct mix between recognizing actual high earners and preventing false positives, which is critical for efficient marketing spending.

4. FEATURE ENGINEERING AND SELECTION

4.1 Feature Engineering: Three new features were engineered from the raw data: Education Ranking, Total Investment Income, Age Grouping. However, upon feature selection using LightGBM importance scores, the model preferred the original raw features age, education, capital_gains, capital_losses and dividends_from_stocks individually as they provided stronger and more direct predictive signals. This demonstrates that tree-based models like LightGBM are often capable of finding their own optimal splits from raw features without requiring manual transformation.

4.2 Feature Selection: Rather than manually deciding which features to keep, a data-driven approach was used. A baseline LightGBM model was trained on all 36 available features and feature importance scores were extracted using the **gain metric** which measures how much each feature contributes to improving the model's predictions.

The results were clear the top 20 features captured **96.8% of total predictive gain** while the bottom 16 features contributed only **3.2%**. Those 16 features were dropped, reducing noise and improving training efficiency without sacrificing performance.



5. TRAINING MODELS

5.1 Baseline Model: Logistic Regression

Logistic Regression was trained as a baseline model to establish a performance benchmark. It works by learning a linear boundary that separates high earners from low earners based on the input features. Despite its simplicity, it serves as a useful reference point. if a more complex model cannot significantly outperform logistic regression, the added complexity is not justified. Class imbalance was handled using the class_weight='balanced' parameter, which automatically adjusts the model to pay more attention to the minority class.

While the baseline model shows promise with a ROC-AUC of 0.93, its precision of 0.50 means every second marketing contact would be wasted, and its recall of 0.56 means nearly half of all high earners are missed entirely. For a retail client where both budget efficiency and audience reach matter, this level of performance is not good enough motivating the move to a more powerful model.

5.2 Primary Model: LightGBM (Optuna Tuned)

Training was done in two stages:
The **first stage** involved training a baseline **LightGBM on all 36 features** purely for feature selection purposes. Feature importance scores from this model were used to identify the top 20 features that contributed 96.8% of the total predictive gain.

The **second stage** involved training the final model on the top **20 features with Optuna hyperparameter tuning**. Optuna uses a smart search strategy called TPE (Tree-structured Parzen Estimator) which learns from previous trials to focus on the most promising parameter combinations. Over 50 trials, it searched across parameters including num_leaves, max_depth, learning_rate, min_data_in_leaf, and regularization terms lambda_l1 and lambda_l2.
Early stopping was used during training, if the validation PR-AUC did not improve for 50 consecutive rounds, training was stopped automatically. This prevents overfitting and saves computation time.

5.2.1 Model Architecture

The final LightGBM used these parameters

- **num_leaves: 226** Controls how many end points each decision tree has.
- **learning_rate: 0.028** How fast or slow the model learns.
- **min_data_in_leaf: 50** Minimum number of people required in each leaf.
- **feature_fraction: 0.527** Only 52% of features are used to build each tree.
- **bagging_fraction: 0.999** Almost all training data is used for each tree.
- **lambda_l1: 0.17 & lambda_l2: 1.58** Regularization
- **min_gain_to_split: 0.73** The model only splits a node if it improves predictions by this amount.
- **scale_pos_weight: 14.84** Tells the model that each high earner is 15x more important than a low earner.

6. MODEL VALIDATIN AND EVALUATION

6.1 Model Validation

Model validation based on a three-way holdout split: 67% training data, 13% validation data, and 20% for final testing. The validation set was only used during the hyperparameter tuning phase (Optuna) to assess each trial while the test set was still a mystery until the final evaluation which was done in an unbiased manner ensuring the closest measure to a real-world performance.
Optuna ran 50 trials using Bayesian optimization, with early stopping applied in each trial halting training if the validation PR-AUC did not improve for 50 consecutive rounds. This is how overfitting was prevented, and unnecessary computing was saved. Validation PR-AUC went up gradually through the trials starting at 0.649 in the beginning trials and reaching a maximum of 0.670 at trial 43 indicating that Optuna search strategy learned from the previous trial to find better parameter sets.

Finally, the completely unseen test set model was evaluated and reached a PR-AUC of 0.686 and ROC-AUC of 0.953, which validated that the model is well prepared for generalization to new data.

6.2 Model Comparison and Evaluation

Since the dataset suffers from an extreme class imbalance, using accuracy as the sole metric can be very deceptive. For instance, a model that labels all individuals as low income would have 94% accuracy, but clearly, it would be of no use at all. So, the classification metrics used for performance evaluation were the following:

PR-AUC: measures the area under the precision-recall curve. This is the most informative metric for imbalanced datasets as it focuses specifically on the minority class performance
ROC-AUC: measures the model's ability to rank high earners above low earners across all thresholds
F1 Score: balances precision and recall, giving a single measure of minority class performance

Precision & Recall: evaluated separately to understand the tradeoff between false positives (wasted marketing spend) and false negatives

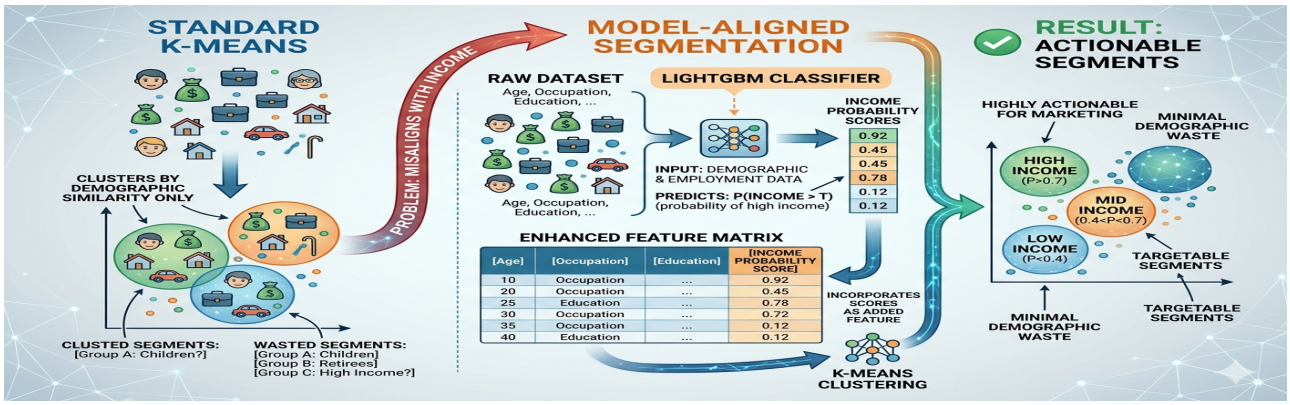
Table 1: Model Performace Comparison

MODELS	ROC-AUC	PR-AUC	Precision	Recall
Logistic Regression	0.927	0.5418	0.4978	0.5578
LightGBM	0.9520	0.6752	0.6074	0.6248
LightGBM OPTUNA TUNED	0.9531	0.6861	0.6489	0.6038

Each of the three models demonstrates an obvious evolution in their capabilities. Logistic Regression set up a pretty decent baseline at PR-AUC of 0.54, however its precision of 0.50 implied that every other marketing contact would be a waste. Transitioning to LightGBM resulted in a remarkable leap, as PR-AUC increased by 24% and precision reached 0.61, thus proving that complexity in income prediction calls for a non-linear model. Optuna tuning took LightGBM to the next level by precision going up to 0.65 and PR-AUC getting to 0.69, that is the model at the end of the line is able to correctly pick out 65 high earners from 100 flagged individuals**. LightGBM Optuna Tuned were the one chosen as the final model** because it performed best in terms of precision and was above par in all the other metrics as well.

SEGMENTATION MODEL FOR USER TARGETING

Standard K-Means clustering groups people purely by demographic similarity, it has no awareness of income. This means it can waste entire clusters on children or retirees rather than finding meaningful high-income segments. To address this, a model-aligned segmentation approach was used, incorporating the LightGBM classifier's income probability scores as an additional clustering feature alongside demographic and employment variables. This guided the clustering towards income-relevant groupings, ensuring the segments are directly actionable for marketing purposes.



1. TRAINING MODELS

13 features were selected for clustering covering demographics, employment, financial activity and the income probability score from the classification model. Three engineered features were created total investment income (combining capital gains, losses and dividends), education rank (ordered numerical ranking of education levels), and the income probability score from the LightGBM model. All categorical columns were label encoded and all features were standardized using StandardScaler to ensure no single feature dominated the distance calculations.

2. PARAMETER SELECTION

Both the elbow method and silhouette score were used to determine the optimal number of clusters. The elbow curve showed no sharp bend, with the most meaningful inertia drops occurring before k=5. Silhouette scores showed

marginal improvements beyond k=5. A value of k=5 was selected as it balances statistical performance with business interpretability five distinct segments are actionable for a retail marketing team without being overly complex.

3. MODEL RESULTS

CLUSTER	NAME	HIGH EARNER %	CUSTOMER TYPE
0	Married non-Workers	0.9%	Bronze Customer
1	High Value Professionals	22.6%	Gold Customers
2	Single Part-timers	1.6%	Silver Customers
3	Children/Dependents	0.0%	Others
4	Ultra Wealthy Professionals	89.4%	Platinum Customers

4. FINDINGS AND EXPLORATION

Several interesting patterns emerged during the data exploration phase that shaped both modeling decisions and business insights.

Class Imbalance: A first and foremost observation was the staggering degree of class imbalance 94% of the population earn less than $50K and only 6% more than that. This was enough to discard accuracy as a meaningful metric and we ended up using PR-AUC and F1 score as the main evaluation measures.

Education: A clear and consistent relationship was found between education and income. Professional degree holders had a 54% high earner rate, doctorate holders 52% and master's degree holders 31% , dropping sharply below 10% for lower education levels.

Age Distribution: Around 28% of the dataset are persons under 18 and only 2 of them are high earners, which makes a rate of 0.003%. So, as expected, age proved to be one of the strongest predictors of income in the dataset. The model very easily picked up on this pattern so there was no need to create the age group.

Investment Income Gap: The investment income gap appeared to be the most extraordinary numerical result that was uncovered. The average investment income for high earners was $6,191 while for low earners it was only $225, that was almost 28 times. Therefore, it was confirmed that wealth produces more wealth with the help of capital gains and dividends.

Working Weeks: High earners are almost working all year round and having an average of 48.1 weeks per year while low earners only have 21.5 weeks which is a bit more than half. Almost half of the set (48%) did not work any weeks which represents mainly children, retirees and unemployed normally.

Wage Per Hour: Despite appearing to be a strong predictor, 94% of the dataset recorded zero hourly wage including most high earners who tend to be salaried, self-employed or living off investments. This made wage per hour a weaker standalone predictor than expected.

5. BUSINESS RECOMMENDATION

The LightGBM income classifier should be used as the first filter in any marketing campaign. Before reaching out to any individual, their demographic and employment information can be passed through the model to generate an income probability score between 0 and 1. Anyone scoring above the optimal threshold of 0.87 should be flagged as a likely high earner and prioritized for targeting.
This approach means the client no longer needs to market to everyone blindly. Instead of a 6% hit rate from random outreach, the model delivers a 65% precision rate, meaning 65 out of every 100 people flagged by the model are genuine high earners. This translates directly into reduced marketing waste and improved return on investment.

Recommended Marketing Strategy per Segment:

Ultra Wealthy Professionals (89.4% high earners): This is the highest priority segment despite its small size. Nearly 9 out of 10 people in this group are confirmed high earners with average investment income exceeding $100,000. The recommended approach is **personalized, premium and exclusive outreach, luxury products, high-end services, VIP offers and white-glove customer experiences**. These are people who are in their middle age, rich and with high educational background, so smart campaigns with products that are luxurious must be advertised to them.

High Value Professionals (22.6% high earners): This is the primary mass marketing target. With nearly 50,000 people and a 22.6% high earner rate, almost 4 times the population average this segment offers the best combination of volume and quality. Marketing should focus on **premium but accessible products**, career and lifestyle-oriented messaging, and channels that reach working professionals such as LinkedIn, professional publications and email campaigns.

Single Part-timers (1.6% high earners): Low-income probability but potentially aspirational audience. Marketing to this group should focus on **value-oriented products, installment payment** options and aspirational messaging. Not recommended for high-income product campaigns but useful for driving volume with mid-range offerings.

Married non-Workers (0.9% high earners): While personal income is low, many in this group file taxes jointly suggesting a working spouse with potentially significant household income. Marketing should focus on household products, family-oriented offerings and lifestyle brands. As majority in this group are females and are middle aged, we can target them with **female oriented products or kids product** as they might have a new born at home. This segment may have more purchasing power than their personal income suggests.

Children & Dependents (0.0% high earners): As this section has majority of kids, this segment should be **excluded from all income-based marketing** campaigns entirely. Resources spent on this group are wasted for the client's stated objective.

6. FUTURE WORK.

Retrain on Recent Data: The current models were trained on 1994-1995 census data. Income patterns, occupational structures and demographic compositions have changed significantly over the past 30 years. Retraining on more recent census data or commercially available demographic datasets would significantly improve the relevance and accuracy of predictions in today's market.

Add Behavioral Data: The current models rely purely on demographic and employment data. Incorporating behavioral signals such as purchase history, browsing patterns, loyalty program data and transaction history would dramatically improve targeting precision. A customer who buys premium products is a stronger signal than demographics alone.

Market Basket Analysis: So currently we are just identifying the rich customers, As future work, the model can be extended to incorporate customer purchase behavior to identify product preferences within each segment, allowing for more targeted recommendations and improved marketing strategies.

Incorporate Ensemble Methods: While LightGBM performed well, combining it with other models such as XGBoost or a Neural Network through ensemble methods could further improve precision. Stacking multiple models often outperforms any single model on complex classification tasks.

7. REFERENCES

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